

HABIB IBRAHIM TOURE

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Computer Science Co-op student at uOttawa — building full-stack and AI-powered systems

EDUCATION

University of Ottawa

Ottawa, ON

B.Sc. (Specialized) in Computer Science, Co-op | International Student Merit Scholarship – \$36,000/yr Sept. 2024 – 2028

- Relevant coursework: Data Structures and Algorithms, Computer Architecture, Introduction to Software Engineering

EXPERIENCE

Data Annotation & AI Training Specialist

June 2025 – Oct. 2025

DataAnnotation

Remote, ON

- Engineered Python automation scripts to eliminate repetitive annotation workflows, reducing manual effort by ~30% across 5+ ML projects
- Evaluated model outputs and delivered structured feedback that measurably improved labeling consistency and model accuracy
- Identified systematic annotation errors in collaboration with a remote AI team, directly refining training data quality guidelines

PROJECTS

MARA | *Python, Claude API, Whisper, PyQt5, Selenium, CUDA, Fish Audio*

Apr. 2026 – Jun. 2026

- Architected a fully local AI voice assistant achieving <1s response latency on simple queries via a parallel Haiku/Sonnet inference pipeline
- Designed hybrid ML pipeline: Whisper (local GPU, CUDA) for speech-to-text, Claude Haiku for intent classification (~150ms), and Claude Sonnet for reasoning — all executing concurrently to eliminate latency bottlenecks
- Implemented autonomous system control (volume, brightness, WiFi, app management) and full browser automation via Selenium with Fernet-encrypted credential storage
- Engineered persistent cross-session memory with automatic conversation summarization, enabling context-aware responses without re-prompting

Fretify | *Python, FastAPI, React, TypeScript, Librosa, Basic-Pitch, Demucs*

Oct. 2025 – Present

- Developing an end-to-end ML application that transforms audio/video recordings into guitar tablature, combining source separation and note detection models
- Architected an audio processing pipeline using Librosa and Basic-Pitch for note detection and MIDI conversion, preceded by Demucs source separation to isolate guitar tracks and reduce detection noise
- Designed full-stack software engineering architecture with a FastAPI backend and React/TypeScript frontend featuring real-time interactive tab visualization

Seren | *React, TypeScript, FastAPI, Claude API, PostgreSQL, Chrome Extension*

Apr. 2026 – Present

- Developing a Chrome extension (Manifest V3) that injects a floating AI toolbar on any text selection, delivering instant Claude-powered explanations, summaries, and adaptive quiz generation
- Architected a full-stack platform with a RESTful FastAPI backend, PostgreSQL database, JWT authentication, and a React dashboard supporting .ics calendar import for automatic deadline syncing
- Engineered a shared JWT session bridge between the extension and web platform, enabling seamless single sign-on with instant state synchronization across both surfaces
- Implemented Pomodoro focus timer and SOS mode in the extension popup to reduce cognitive load, displaying one prioritized task at a time during high-stress periods

TECHNICAL SKILLS

Languages: Python, Java, JavaScript, TypeScript, HTML/CSS, SQL

Frameworks & Libraries: React, FastAPI, Node.js, SQLAlchemy, PyTorch, Scikit-learn, NumPy, Librosa, Selenium, PyQt5

Databases: PostgreSQL, MySQL, SQLite

Developer Tools: Git, Docker, Linux, REST APIs, Chrome Extension (MV3), JWT, Agile, CI/CD

Other: Bilingual (English/French), Machine Learning Engineering, CUDA / GPU Inference, API Development